Microeconomics I

Class 9 — Uncertainty II: risk aversion measured; stochastic dominance

Optional reading. Not taught in class and **not examined in the Part I exam** (Wednesday 28 October 2026).
It is part of the material for the final exam (Friday 18 December 2026).

Arrow–Pratt measures · comparing risk aversion · insurance · first- and second-order stochastic dominance · mean-preserving spreads

Mas-Colell, Whinston and Green (MWG), sections 6.C–6.D · each exercise: the question, then its answers on the next page

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Universitat de Barcelona · Master in Economics · 27 October 2026

Valencia, October 2024: how much did the flood insurer pay?

29 October – 4 November 2024: floods in Valencia
Spain's public Consorcio pays the flood damage to insured homes, cars and firms.

How much had it paid by November 2025?

40 · 400 · 4,000 million euros

Consorcio de Compensación de Seguros, *Consorseguros Digital* no. 23 (autumn 2025), "La DANA de Valencia 2024"; figures as of 24 November 2025.

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more than 4,000 million euros by 24 November 2025: its largest in over 70 years
Paid by a surcharge included in private home, car, life and business policies.

Consorcio de Compensación de Seguros, *Consorseguros Digital* no. 23 (autumn 2025), "La DANA de Valencia 2024"; figures as of 24 November 2025.

Concave u puts the certainty equivalent 9 euros below the mean

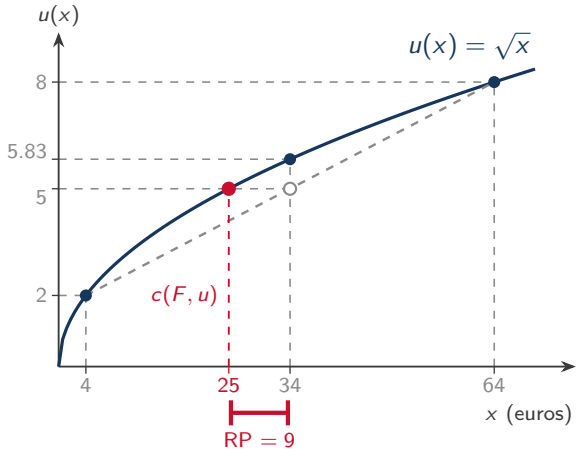
Certainty equivalent $c(F, u)$:

$u\Big(c(F,u)\Big) = \int u(x)\, dF(x).$

Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

The chord lies below the curve:
$c(F,u) < E[x]$.



Class 8, slides 19–21: $u = \sqrt{x}$; 4 or 64 euros, $\frac{1}{2}$ each. MWG Def. 6.C.2(i), p. 186; Prop. 6.C.1, p. 187.

Rescale u by 10: u'' is multiplied by 10, the choices are the same

Class 8: the Bernoulli $u = 10\sqrt{x} + 3$ ranks lotteries as $\sqrt{x}$ does.

	$u = \sqrt{x}$	$\tilde{u} = 10\sqrt{x} + 3$
$u'(25)$	0.1	1
$u''(25)$	-0.002	-0.02
$-u''(25)/u'(25)$	0.02	0.02
$c(F, \cdot)$, 4 or 64 euros	25 euros	25 euros

u'' changes with the scale of u; the ratio $-u''/u'$ does not.

MWG 6.C, p. 190: u'' "is not invariant to positive linear transformations"; Fig. 6.C.4.

Arrow–Pratt: $r_A(x) = -u''(x)/u'(x)$ and $r_R(x) = x\, r_A(x)$

$$r_A(x) = -\frac{u''(x)}{u'(x)}$$

absolute risk aversion (risks in euros)

$$r_R(x) = -\frac{x\, u''(x)}{u'(x)} = x\, r_A(x)$$

relative risk aversion (risks as shares of wealth)

$$u = \sqrt{x}\text{:}\quad r_A(x) = \frac{1}{2x},\quad r_R(x) = \tfrac{1}{2}$$

r_A falls with x (class 8's Ana, $\sqrt{x}$, 64 euros if two coins both land heads):
RP 12 euros on the game alone, 3 euros with 36 sure euros more

Both unchanged by $\beta u + \gamma$, $\beta > 0$.

MWG Def. 6.C.3, p. 190; Def. 6.C.5, p. 194; decreasing absolute risk aversion: Def. 6.C.4, p. 193.

Exercise 1: r_A, r_R and the attitude of four Bernoulli functions

Three answers per row ($x > 0$). The answers are on the next page.

	$u(x)$	$r_A(x)$	$r_R(x)$	attitude
(a)	$4\sqrt{x} + 1$			
(b)	$-e^{-x/50}$			
(c)	$-\dfrac{1}{x^2}$			
(d)	x^2			

MWG Example 6.C.4 (constant r_A), p. 191; Def. 6.C.5, p. 194.

Exercise 1: r_A, r_R and the attitude of four Bernoulli functions

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	$u(x)$	$r_A(x)$	$r_R(x)$	attitude
(a)	$4\sqrt{x}+1$	$\frac{1}{2x}$	$\frac{1}{2}$	averse
(b)	$-e^{-x/50}$	$\frac{1}{50}$	$\frac{x}{50}$	averse
(c)	$-\frac{1}{x^2}$	$\frac{3}{x}$	3	averse
(d)	x^2	$-\frac{1}{x}$	-1	loving

(a): as $\sqrt{x}$. (b): constant absolute, (c): constant relative risk aversion (CARA, CRRA).

MWG Example 6.C.4 (constant r_A), p. 191; Def. 6.C.5, p. 194.

More risk averse: r_A never lower, certainty equivalents never higher

u_2 is more risk averse than u_1 — three equivalent tests:

(i) $r_A(x, u_2) \geq r_A(x, u_1)$ for every x — at least as curved everywhere

(ii) $u_2 = \psi(u_1)$, ψ increasing and concave

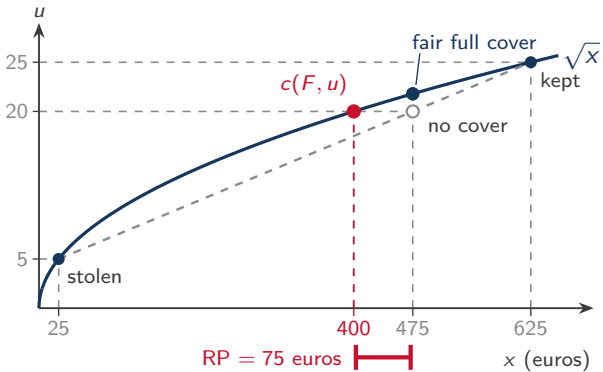
(iii) $c(F, u_2) \leq c(F, u_1)$ for every F — in euros

A partial order: many pairs are not comparable.

Compare people by certainty equivalents, never by utility numbers.

MWG Prop. 6.C.2, p. 191: (i)–(v) are equivalent; (iv) and (v) are not needed here. Partial ordering, p. 192.

Insurance at a fair price: full cover



a 600-euro bike + 25 euros

stolen with probability $\pi = \frac{1}{4}$

no cover: 625 or 25 euros

$E = 475$, $c = 400$ euros

fair cover, $q = \pi$ per euro:
600 euros of cover cost 150

475 euros for sure

At a fair price every strictly risk-averse buyer insures fully (cover = the 600-euro loss). She would pay up to 225 euros for an expected payout of 150 euros.

MWG Example 6.C.1, pp. 187–188; price above π: partial cover, p. 189.

Exercise 2: who is more risk averse, Bo or Cy?

Bo and Cy maximize expected utility over wealth $x > 0$ (euros).
Bo: $u_B(x) = -1/x$. Cy: $u_C(x) = x^{1/3}$.

(a) r_A and r_R for Bo and for Cy.

(b) Is one of them more risk averse (larger r_A at every x)?

(c) As wealth rises, does each r_A rise, fall or stay the same? And r_R?

MWG Def. 6.C.3, p. 190; Prop. 6.C.2, p. 191; Def. 6.C.4, p. 193; Def. 6.C.5, p. 194.

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Bo and Cy maximize expected utility over wealth $x > 0$ (euros).
Bo: $u_B(x) = -1/x$. Cy: $u_C(x) = x^{1/3}$.

(a) r_A and r_R for Bo and for Cy.

Bo: $r_A = 2/x$, $r_R = 2$. Cy: $r_A = 2/(3x)$, $r_R = \frac{2}{3}$.

(b) Is one of them more risk averse (larger r_A at every x)?

Bo: $2/x > 2/(3x)$ at every $x > 0$. At 50 euros: 0.04 against 0.013.

(c) As wealth rises, does each r_A rise, fall or stay the same? And r_R?

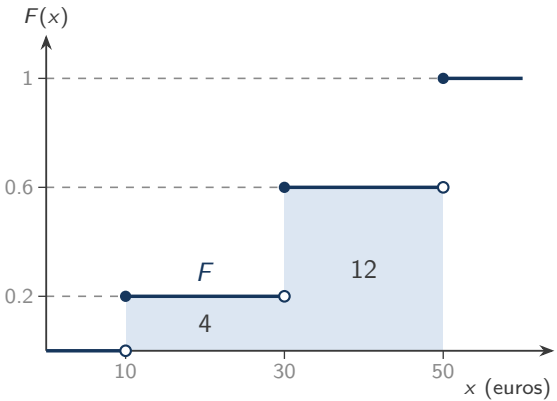
Both r_A fall with wealth (decreasing absolute risk aversion). Both r_R stay constant.

MWG Def. 6.C.3, p. 190; Prop. 6.C.2, p. 191; Def. 6.C.4, p. 193; Def. 6.C.5, p. 194.

Comparing lotteries: stochastic dominance

MWG 6.D

A lottery's CDF: $F(x)$ = Prob(at most x); jumps are probabilities



F: 10, 30, 50 euros
with 0.2, 0.4, 0.4

$F(x) = \text{Prob}(\text{payoff} \leq x)$

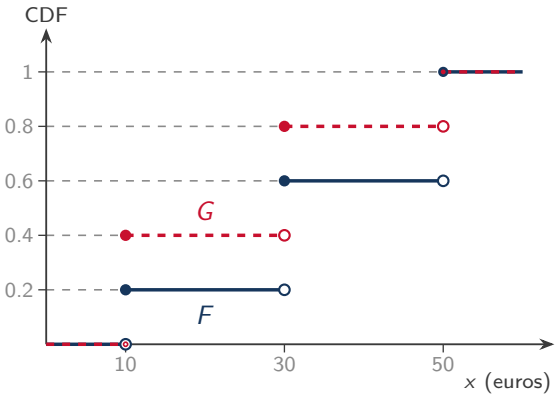
jump at 30 euros: $0.6 - 0.2 = 0.4$

area under F up to 50 euros:
$4 + 12 = 16$ euros
$= 50 - E[x] = 50 - 34$

Equal means $\Leftrightarrow$ equal areas under the two CDFs (from 0 to where both reach 1).

MWG 6.C, pp. 183–184, Fig. 6.C.1; (6.D.1), p. 198.

F dominates G (first order): $F(x) \leq G(x)$ at every x



first-order stochastic dominance (FOSD)
F FOSD G: $\int u\,dF \geq \int u\,dG$ for every nondecreasing u

everyone who likes money agrees

$\Leftrightarrow$ $F(x) \leq G(x)$ for every x

on	$[10, 30)$	$[30, 50)$	$[50, \infty)$
F	0.2	0.6	1
G	0.4	0.8	1

F never above G: everyone who likes money weakly prefers F (means 34, 26 euros).

MWG Def. 6.D.1 and Prop. 6.D.1, p. 195; Fig. 6.D.1, p. 196. G: 10, 30, 50 euros with 0.4, 0.4, 0.2.

Exercise 3: two CDFs, P and Q

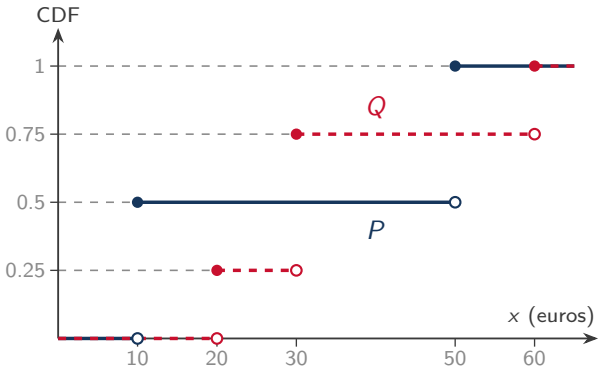
P: 10 or 50 euros, $\frac{1}{2}$ each Q: 20, 30, 60 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$

Draw both on one graph; compare interval by interval. Does either dominate? Answers on the next page.

Exercise 3: two CDFs, P and Q

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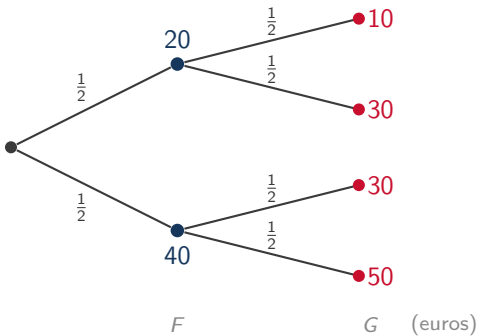
neither

P above Q on $[10, 30)$ and $[50, 60)$;
Q above P on $[30, 50)$

$E_P = 30 < E_Q = 35$ euros

yet P is preferred by
$u = 1$ from 50 euros (else 0)

Same mean, more spread: every risk averter weakly prefers F



second-order stochastic dominance (SOSD)
F SOSD G: same mean, and $\int u\,dF \geq \int u\,dG$ for every nondecreasing concave u

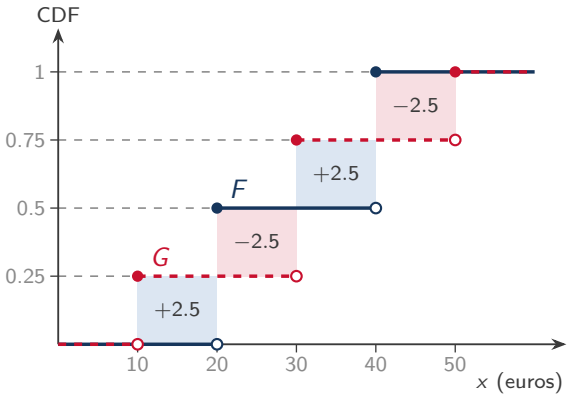
mean-preserving spread: each outcome of F becomes a lottery with that mean

G: 10, 30, 50 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$.
Both means: 30 euros.

G is a mean-preserving spread of F, so F SOSD G.

MWG Def. 6.D.2, p. 197; Example 6.D.2, pp. 197–198.

Same mean, $\int_0^x G \geq \int_0^x F$ at every x: F dominates G (SOSD)



$\int_0^x G(t)\,dt \geq \int_0^x F(t)\,dt$ for every x

MWG (6.D.2): areas under the CDFs, running from 0

x	10	20	30	40	≥ 50
$\int_0^x (G-F)$ (euros)	0	2.5	0	2.5	0

Never negative, and 0 from 50 euros on (equal means): F SOSD G; G is a mean-preserving spread of F.

MWG (6.D.1)–(6.D.2), p. 198; Prop. 6.D.2, p. 199: (i) SOSD $\Leftrightarrow$ (ii) mean-preserving spread $\Leftrightarrow$ (iii) (6.D.2).

Exercise 4: which bonus is less risky? Areas under the CDFs

F pays 20 or 60 euros, probability $\frac{1}{2}$ each.
G pays 10, 30, 40 or 80 euros, probability $\frac{1}{4}$ each.
F and G also denote their CDFs.

(a) Are the two means equal?

(b) Compute $\int_0^t F(s)\,ds$ and $\int_0^t G(s)\,ds$ at $t = 10, 20, 30, 40, 60, 80$. Is $\int_0^t G \geq \int_0^t F$ at every t, so that F SOSD G?

(c) With $u(x) = -1/x$: $c(F, u)$, and $\mathrm{RP} = E[x] - c(F, u)$ for F.

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(c) With $u(x) = -1/x$: $c(F, u)$, and $\mathrm{RP} = E[x] - c(F, u)$ for F.

(a) Both means are 40 euros.

(b) $\int F$: 0, 0, 5, 10, 20, 40. $\int G$: 0, 2.5, 5, 10, 25, 40. Both integrals are linear between these points, so $\int G \geq \int F$ at every t: F SOSD G.

(c) $E[u] = -\frac{1}{2}(\frac{1}{20} + \frac{1}{60}) = -\frac{1}{30}$ utils, so $c(F, u) = 30$ euros and $\mathrm{RP} = 40 - 30 = 10$ euros.